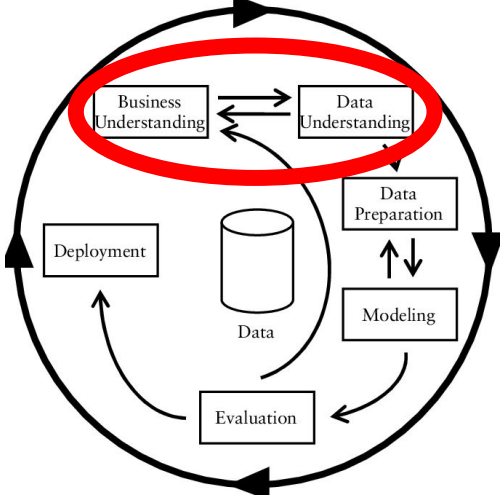
CSC 5793-4933 Final Project

Student	Role
Kiefer Armand	Project Lead
William Becker	Preprocessing/Mappings
Reese Haydel	Modeling/Validation
Keahi Temple	Modeling/Validation

Presentation/Project Overview



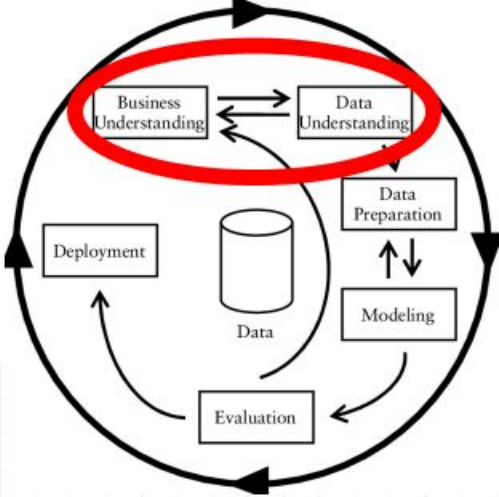
<u>Objective:</u> anomaly detection framework

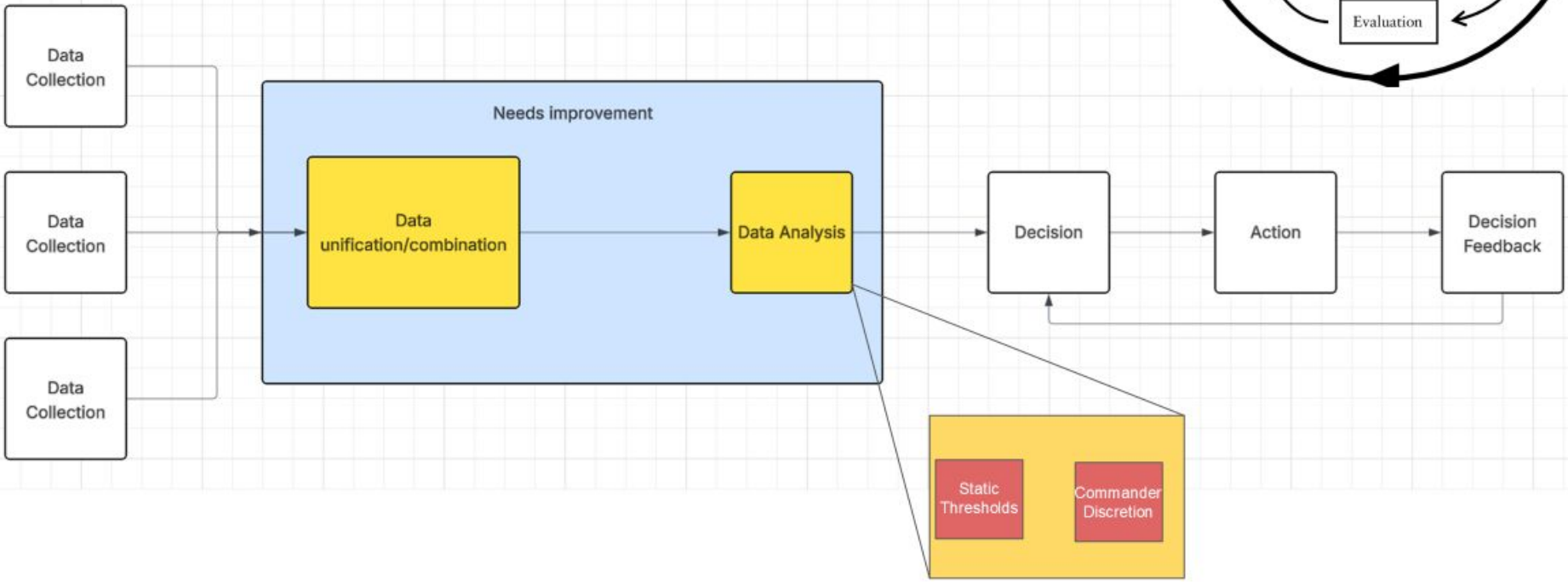
<u>DoDAF System:</u> Command and Control (C2) systems

<u>Current State:</u> Fixed, predefined thresholds to detect abnormal behavior or trigger alerts.

Dataset: UAVSwarm

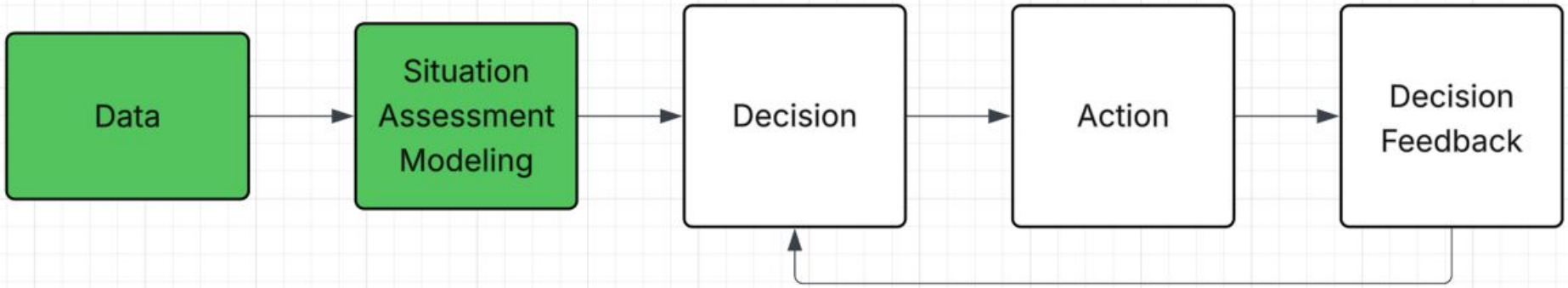
System Boundaries



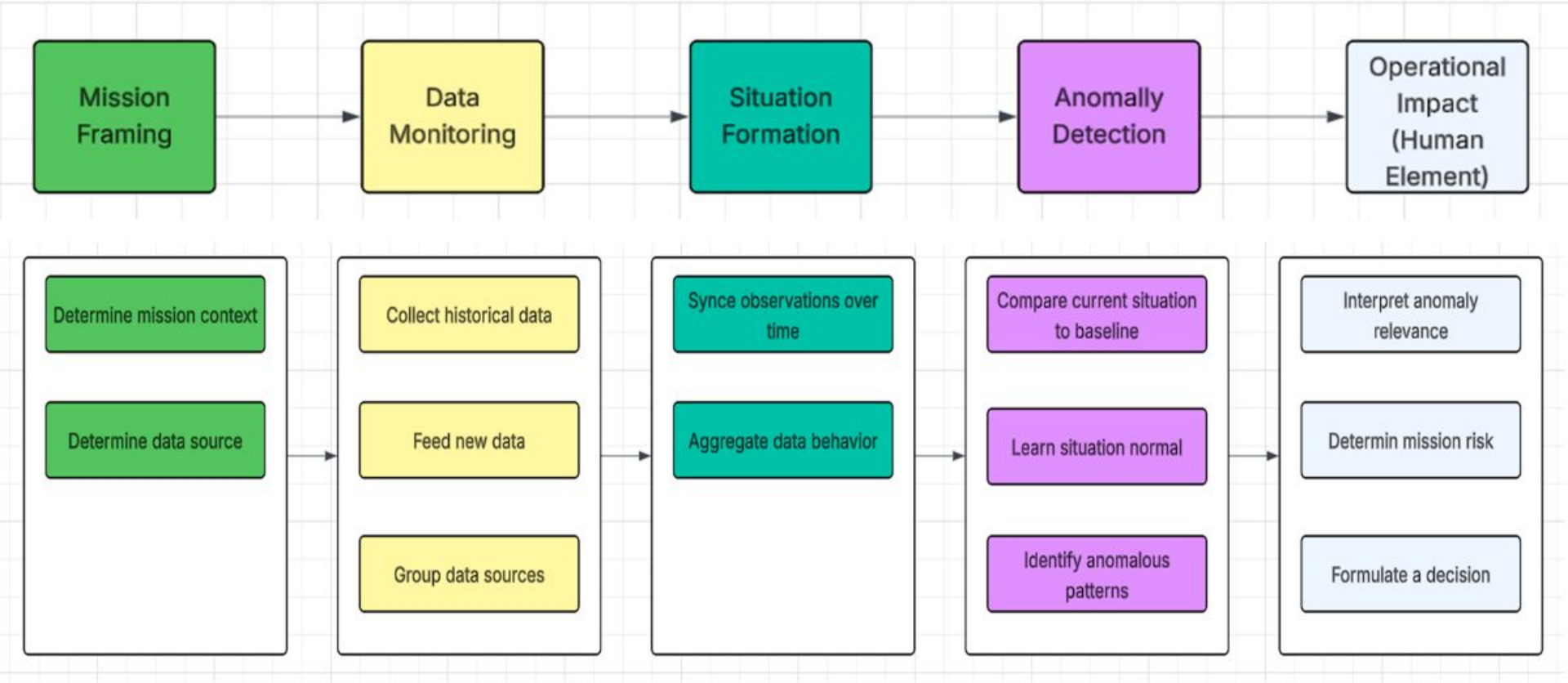


Feature	Description	Datatype	Example
Frame	Time index representing the video frame or timestep in which the detection occurred	Integer	1
X	Horizontal pixel coordinate of the top-left corner of the detected bounding box	Integer	117
Y	Vertical pixel coordinate of the top-left corner of the detected bounding box	Integer	145
Width	Width of the detected bounding box in pixels	Integer	31
Height	Height of the detected bounding box in pixels	Integer	25
Confidence	Detection confidence score from the object detection model	Float	1.0

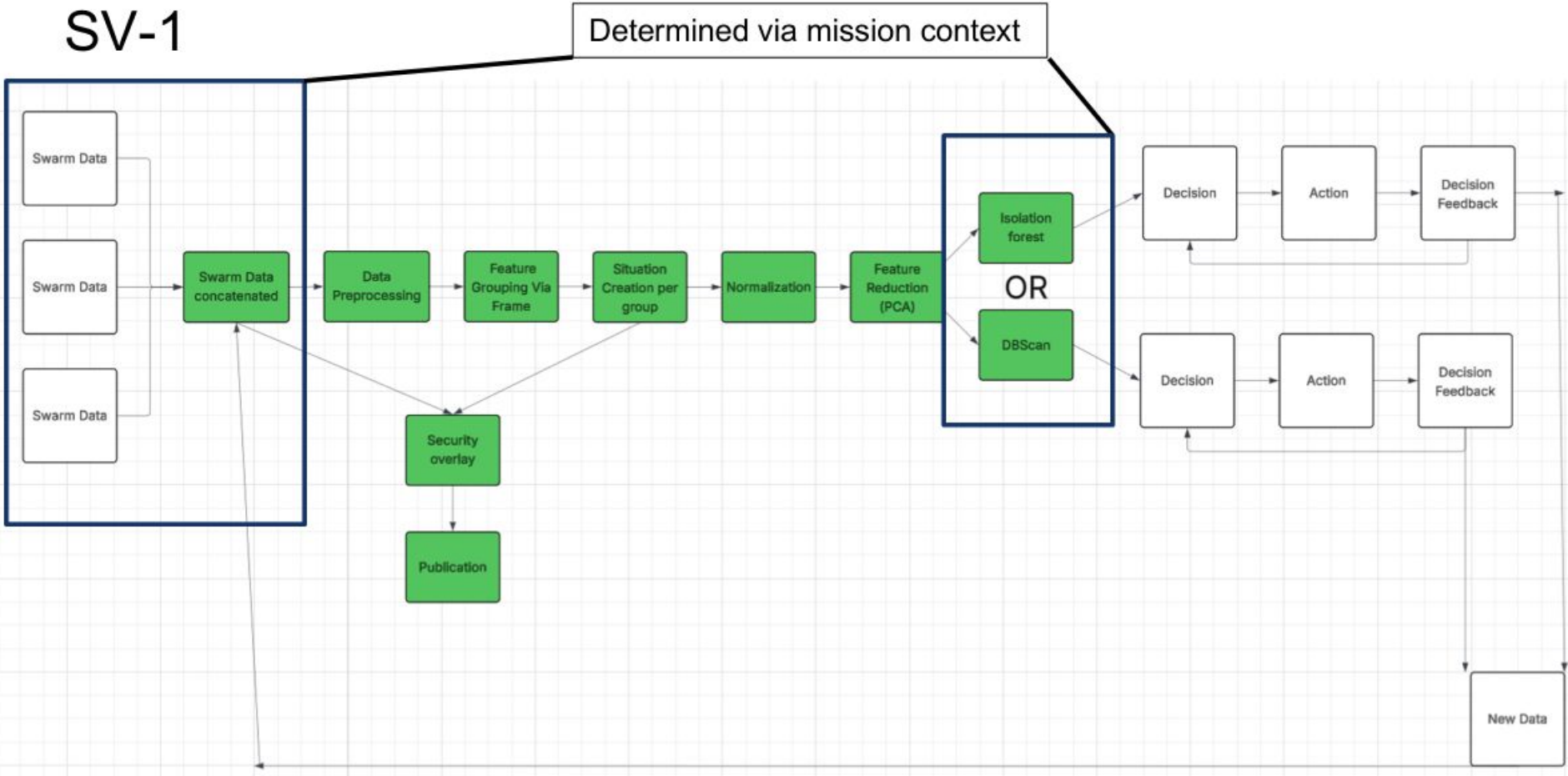
Project Scope – CONOPS/OV-1



OV-2 / OV -5


Mission Framing
Data Monitoring
Situation Formation
Anomally Detection
Operational Impact (Human Element)
Determine mission context
Determine data source
Collect historical data
Feed new data
Group data sources
Synce observations over time
Aggregate data behavior
Compare current situation to baseline
Learn situation normal
Identify anomalous patterns
Interpret anomaly relevance
Determin mission risk
Formulate a decision

SV-1


Determined via mission context
Swarm Data
Swarm Data
Swarm Data
Swarm Data concatenated
Data Preprocessing
Feature Grouping Via Frame
Situation Creation per group
Normalization
Feature Reduction (PCA)
Isolation forest
OR
DBScan
Decision
Action
Decision Feedback
Decision
Action
Decision Feedback
Security overlay
Publication
New Data

SV-6

Producing System	Consuming System	Data Element In	Data element Out
Swarm data source	Data Concatenation System	Individual detection sets	Combined detection data (base dataset)
Data Concatenation System	Data Preprocessing	Combined detection data (base dataset)	Cleaned detection data
Data Preprocessing	Frame Grouping	Cleaned detection data	Frame grouped data
Frame grouping	Situation creation	Frame grouped data	Feature engineered Situation dataset
Situation Creation	Normalization	Feature engineered Situation dataset	Normalized data
Normalization	Feature Reduction	Normalized data	PCA data
Feature Reduction	Data Molding	PCA data	Dataset anomalies
Data Modeling	C2 interface (Human)	Dataset anomalies	Mission decision

Traceability

OV	SV Mapping
Mission Framing	C2 Interface (Human)
Data Monitoring	Swarm data, Data Concatenation, Data Preprocessing
Situation Formation	Frame grouping, Situation Creation
Anomaly Detection	Feature reduction, Data Modeling
Operational Impact	C2 Interface (Human)

T. Ghafouri and J. Shahrabi, "Optimal Thresholds for Anomaly-Based Intrusion Detection in Dynamical Environments," *arXiv:1606.06707*, 2016.

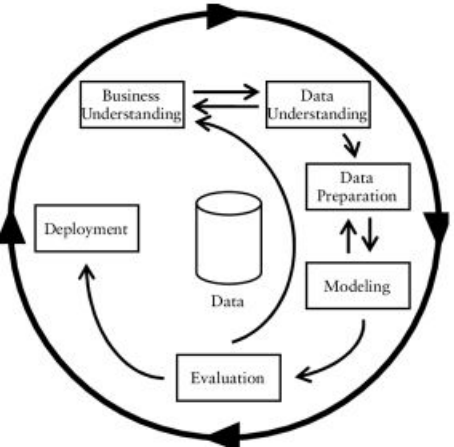
R. A. Bridges, R. M. Vaughn, and G. Cybenko, "Setting the Threshold for High-Throughput Detectors," *arXiv:1710.09422*, 2017.

Version Tracker

Version	What Changed	Result
v0.1	Applied data exploration/some preprocessing	need for feature engineering to match project focus
v0.2	Feature Engineering/Normalization/Modeling	output model that detects situation anomalies
v0.3	Model validation and Security	Model ready to deploy
v0.4	Deployment	Model deployed

Project Pivots

Date	Decision / Change	Reason	Impact
1/26/2026	Selected Command and Control, and UAVSwarm	Room for improvement, while still easy to understand	Project focus/direction
2/5/2026	Used CripDM cycle Business Understanding ->Data Understanding to refine our project focus (see project overview)	Further project details, and analysis of data	Streamlined project plan
2/11/2026	Use Unsupervized ML for detecting situation anomolies	Anomaly detection fits our business understanding better	Selected Models for data
2/22/2026	Business Understanding of project modified as a direct result of model output	Both models presented as viable options, hence business understanding was changed to adapt to each model depending on missin parameters	Project has moved forward into deployment phase



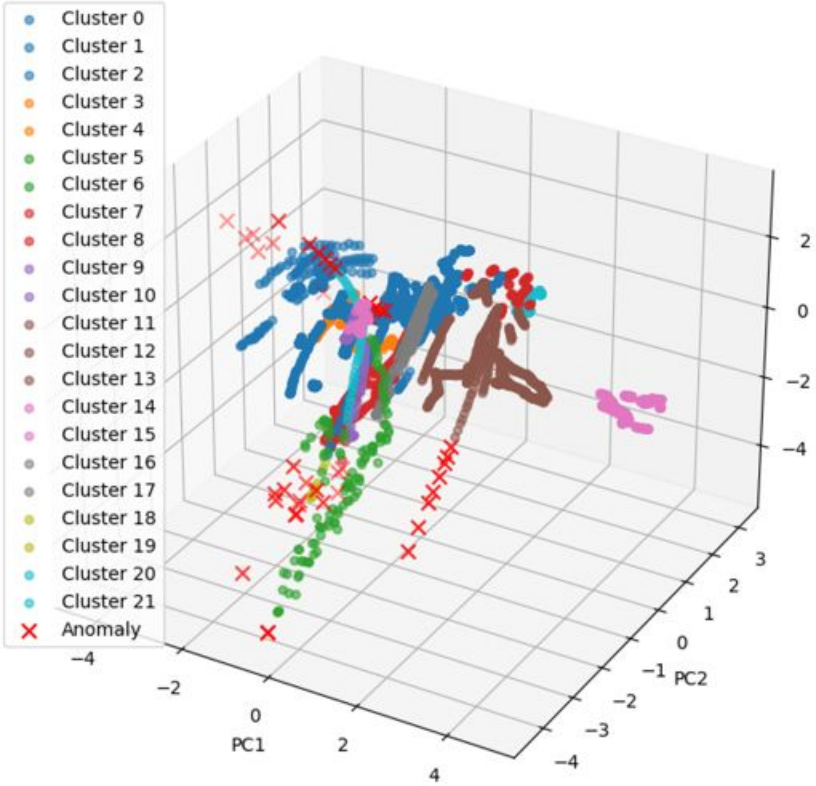
Blockers

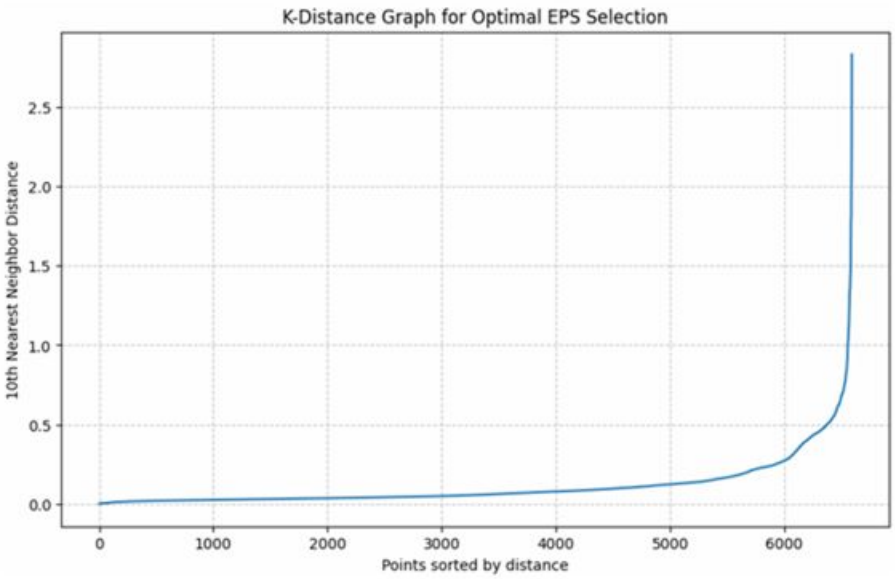
Blocker	Risk	Mitigation
Time Conflicts	Project Delays	Shift responsibilities to assist when one group member is unable to finish a task on time
Security Overlay	Reversable data published	Mission/Project Framing

Security

Dataset	Risk Level	Reasoning
Detection (base dataset)	High	Raw positional data, full capability to reconstruct
Feature Engineered	Moderate	Spatial Structure still partially present
PCA	Low	Only PCA-transformed data (no loadings)
Train/Test Split	Low	Only PCA-transformed data (no loadings)

Results - DBSCAN

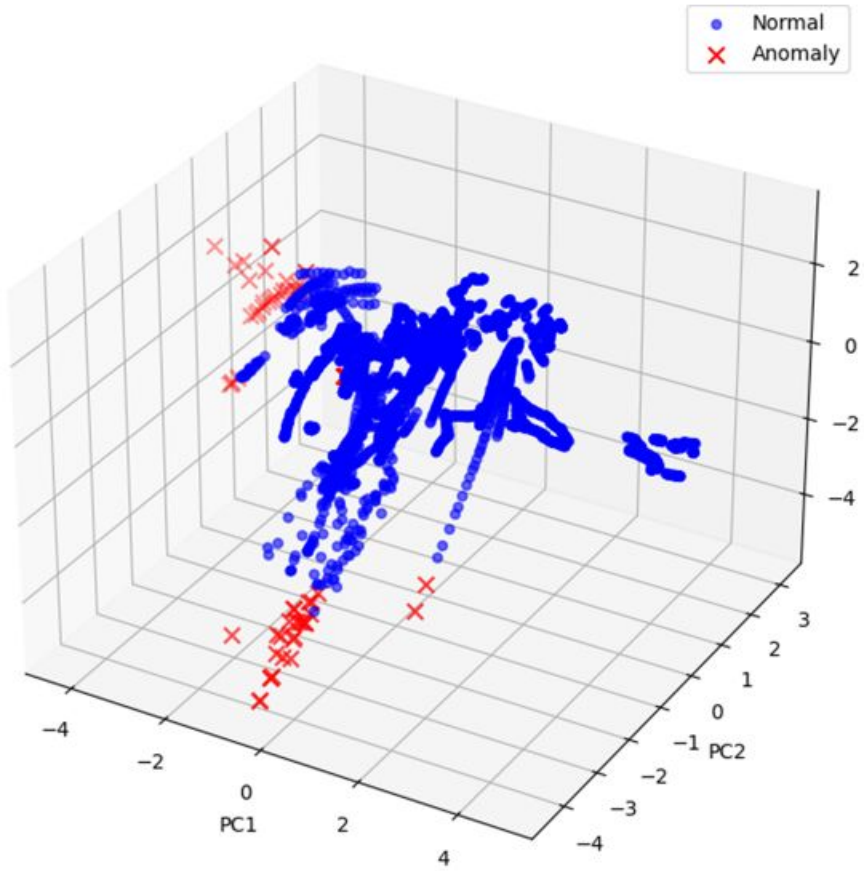




Scoring Summary:

- Minimum Distance to CP: 0.6009
- Maximum Distance to CP: 3.8345
- Mean Distance: 1.3705

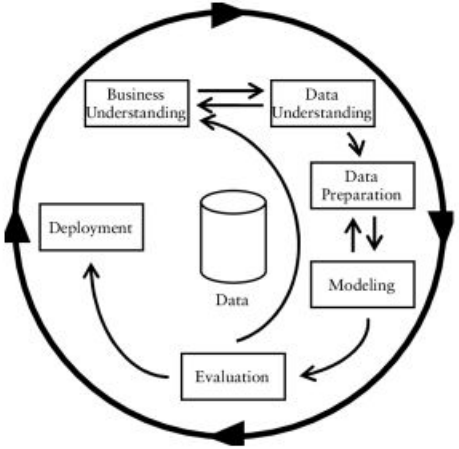
Results - Isolation

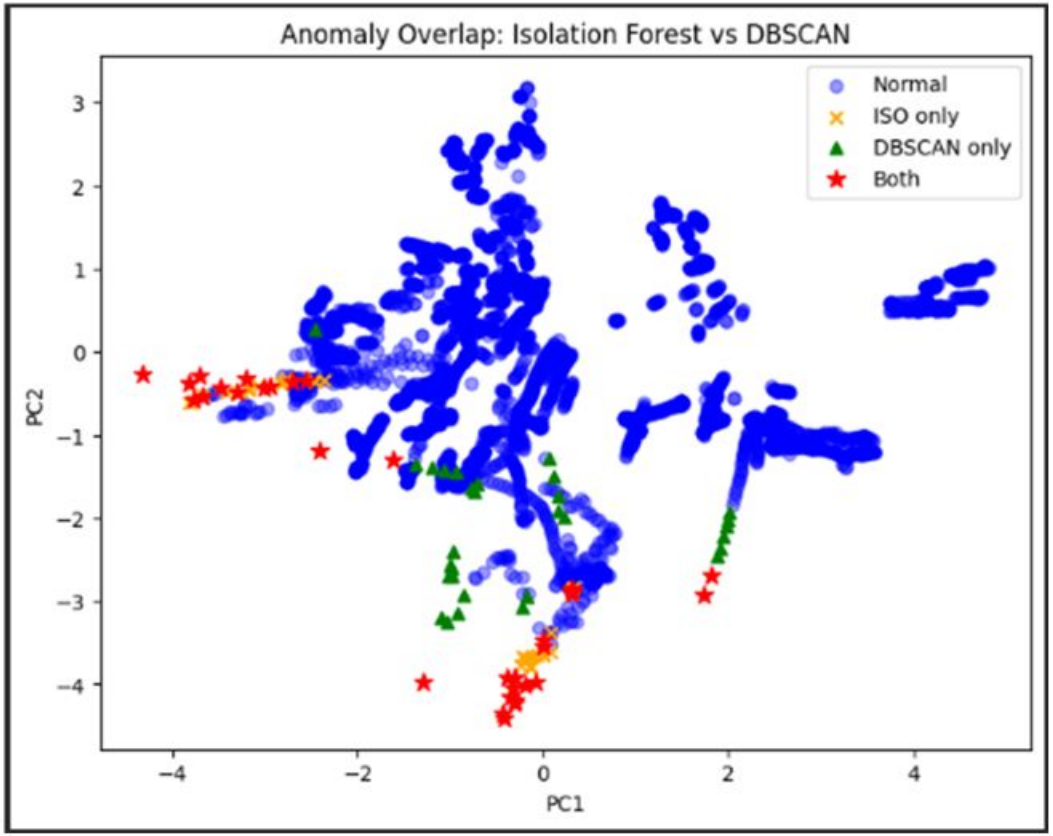



```
# Most anomalous rows in TRAIN
train_anom_view = train_out.sort_values("if_score_samples", ascending=True).head(25)
train_anom_view
```

	PC1	PC2	PC3	PC4	PC5	if_pred	if_score_samples	if_decision
3607	-4.378196	-0.282873	1.934088	-4.262204	3.294898	-1	-0.704613	-0.143701
542	-0.423217	-4.404636	-5.170278	2.066299	0.601634	-1	-0.694172	-0.133260
2568	-2.436767	-1.204407	3.194647	-2.444393	2.935742	-1	-0.693927	-0.133015
1922	-0.311718	-4.228051	-4.677632	2.174412	0.344592	-1	-0.693583	-0.132670
3473	-0.361742	-4.157663	-4.733920	2.050292	0.459921	-1	-0.690875	-0.129963
2863	-0.304186	-4.189359	-4.661774	1.937621	0.518846	-1	-0.687237	-0.126325
1	-0.200170	-4.017518	-4.261421	2.091311	0.208783	-1	-0.686381	-0.125469
3973	-3.515104	-0.438353	1.463056	-3.201707	2.154917	-1	-0.684119	-0.123206
3784	-1.304748	-3.975596	-4.160111	1.671571	0.861916	-1	-0.683950	-0.123038
2130	-0.310747	-3.917474	-3.841306	2.011851	0.249879	-1	-0.678711	-0.117799
2486	-0.077900	-3.969703	-4.292203	1.916785	0.370202	-1	-0.678704	-0.117791
5009	-0.390583	-3.915769	-3.835317	1.993238	0.283302	-1	-0.677031	-0.116119
3518	-0.153218	-3.789924	-3.967033	1.902531	0.228339	-1	-0.671901	-0.110989
3500	-2.800118	-0.319047	1.606917	-2.943931	1.808937	-1	-0.661601	-0.100688
1536	-0.128730	-3.796739	-3.895408	1.680689	0.381942	-1	-0.661443	-0.100531
4525	-1.638127	-1.312247	2.900726	-1.589640	2.038857	-1	-0.657014	-0.096101
737	-0.228190	-3.668665	-3.306363	1.872027	0.143882	-1	-0.656931	-0.096019
1802	-0.197770	-3.677608	-3.295599	1.856563	0.167039	-1	-0.654300	-0.093388
1380	1.745199	-2.933489	-3.238923	0.705624	1.406833	-1	-0.651909	-0.090997
1274	-0.151327	-3.693613	-3.588633	1.683710	0.302861	-1	-0.650925	-0.090013
3121	-0.106313	-3.675471	-3.603429	1.656250	0.298812	-1	-0.650418	-0.089506
1180	-2.840695	-0.356119	1.369153	-2.712341	1.614780	-1	-0.649132	-0.088220
4874	-2.512831	-0.353904	1.609415	-2.732437	1.638633	-1	-0.646061	-0.085148
1202	-3.206630	-0.445439	1.007101	-2.685285	1.540348	-1	-0.646054	-0.085142
5223	-0.003844	-3.652235	-3.613004	1.582508	0.318170	-1	-0.644665	-0.083753

Overlap


Business Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment
Data


Anomaly Overlap: Isolation Forest vs DBSCAN
Normal
ISO only
DBSCAN only
Both
PC1
PC2
−4
−2
0
2
4
3
2
1
0
−1
−2
−3
−4

Future Improvements

- Balance Anomaly scoring with Cluster analysis
- Integrate and automate the decision feedback loop
- Add the ability to flag specific behavior